ECO358 – Assignment 2

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1 Executive summary

In this report, I construct and value a portfolio of four Canadian TSX firms: Enbridge Inc. (ENB), Alimentation Couche-Tard (ATD), Magna International (MG), and Cameco Corporation (CCO) with the benchmark is the TSX Composite Index, over a five-year historical period (2021-2026). The model applied is Capital Asset Pricing Model (CAPM) throughout the whole report. The market risk premium is estimated at 9.24%, market volatility is 12.91%, and the risk-free rate is 3.03% (as of March 11, 2026).

After the analysis with CAPM, I computed the bete for each stock as ENB (β = 0.63) and ATD (β = 0.71) have low risk sensitivity, while MG (β = 1.35) and CCO (β = 1.79) have higher market risk. The implied returns are 8.89%, 9.63%, 15.46%, and 19.57% for ENB, ATD, MG, and CCO. However, realized returns differ a lot from CAPM predictions. ENB (17.21%) and ATD (17.57%) show positive alpha while CCO's very high return (53.09%) leads to a large positive alpha, likely reflecting strong outperformance in the uranium sector. By contrast, MG's return (1.39%) is far below its CAPM estimate, reflecting a deeply negative excess return.

The tangent portfolio is ENB 57%, ATD 20%, CCO 24%, and MG 0 %. MG receives zero weight because its very low return and high risk make it less competitive compared to the other stocks. The portfolio delivers an annual return of 25.73%, volatility of 17.27%, and a Sharpe ratio of 1.31, which is much higher than the TSX benchmark (about 0.72), a proof that the tangent portfolio provides a better return for each unit of risk than the market index.

In Security Market Line (SML) plot, all stocks have clear deviations from SML. In short, ENB, ATD, and CCO are all above the SML (positive alpha), while MG is far below (negative alpha). Under CAPM's assumptions of investors' rational behavrious, competitive and efficient market, such deviations imply that some of them have failed in practice. The real situations are reflecting market inefficiencies, behavioral biases, or even a need for a more complicated model to better capture investing strategies and measure risk instead of relying soly on alpha, beta, or Sharpe ratio metrics. All data, analysis and computations are in this Excel file ([1]).

2 Firm and Industry

Enbridge (ENB.TO)

Enbridge Inc. is Canada's largest energy infrastructure company, operating the world's longest crude oil and liquids transportation system across Canada and the U.S. The company runs four main business lines: liquids pipelines, gas transmission, gas distribution, and renewable power.

Enbridge is a strong addition to my portfolio because it provides indirect exposure to the oil sector as its revenues depend more on energy transportation demand than taking the risks on volatile oil prices.

Alimentation Couche-Tard Inc. (ATD.TO)

Alimentation Couche-Tard is a Canadian multinational company that operates convenience stores and fuel stations, with major brands like Circle K.

Couche-Tard is a good addition because it offers relatively steady cash flows due to consistent consumer demand for fuel and everyday goods. The company has grown a lot by buying other companies, so it's well diversified and not too dependent on one market.

Cameco Corporation (CCO.TO)

Cameco Corporation is the world's largest publicly traded uranium company, based in Saskatoon, Saskatchewan, Canada. Even though CCO has highest return, as well as highest risk, it is a strong addition to my portfolio because of the soaked demand of nuclear power and the market is also having a deficit in uranium supply-demand ([2]).

Magna International Inc. (MG.TO)

Magna International is a global automotive supplier operating through four segments: Body Exteriors & Structures, Power & Vision, Seating Systems, and Complete Vehicles, with operations across North America, Europe, and Asia Pacific. The company is facing a slowdown in production because of tarrif cost, EV transition uncertainty, low demand, and supply-chain disruptions.

3 Portfolio Construction

I chose the four firms to represent four distinct sectors of the Canadian economy — energy infrastructure (ENB), convenience retail (ATD), automotive manufacturing (MG), and uranium mining (CCO), with a goal of having a cross-sector selection to follow a diversification principle: companies in certain industries may outperform those in industries facing different economic conditions or stages of the business cycle, so holding stocks from different sectors reduces the simultaneous movements across all securities.

The correlation matrix shows how each firm's returns move in relation to one another. All six pairwise correlations are low, ranging from 0.153 (ATD-CCO) to 0.221 (ENB-ATD). The number suggests that the stocks respond to different economic drivers, and this benefits the diversification, as if one is doing poorly, the others aren't automatically dragged down.

4 Question 1

Risk-free rate

I chose the risk-free rate of the 5-year Government of Canada bond as the benchmark because the issuing Government for all 3 firms is Canada, and the Government bond is considered free of default risk. And because the maturity of risk-free rate must match the time horizon of the cash flows with data from 03/2021 to 03/2026 (p. 449, [3]), I picked a 5-year time horizon which is long-term and adequate for valuing equity investments.

The risk-free obtained from the Bank of Canada as of March 11, 2026 is 3.03% [4].

1a. Historical daily prices for each stock and for the index

Historical daily adjusted closing prices for Enbridge Inc. (ENB), Alimentation Couche-Tard Inc. (ATD), Cameco Corporation (CCO), and Magna International Inc. (MG) were downloaded from Yahoo Finance for the period March 11, 2021 to March 10, 2026, and there was approximately 1,254 trading days (as shown in Figure 1).

The S&P/TSX Composite Index (GSPTSE) was chosen as the market proxy, because all four firms are listed on the Toronto Stock Exchange.

Adjusted closing prices were used rather than raw closing prices, as they account for dividends and stock splits, ensuring that computed returns reflect total returns to investors.

1b. Daily returns

To compute the return for each day for all stocks and index TSX, I use the formula:

$$R_{\mathrm{i},t} = \frac{P_t - P_{t-1}}{P_{t-1}} \tag{1}$$

Date	ENB	ATD	MG	CCO	TSX
2021-03-11	32.91	39.27	96.54	20.47	18844.6
2021-03-12	32.85	39.83	96.76	21.25	18851.3
2021-03-15	32.99	39.76	97.41	22.92	18954.8
2021-03-16	32.79	40.03	95.12	22.97	18874
2021-03-17	32.87	39.71	97.67	23.61	18983.1
2021-03-18	32.43	38.89	95.94	22.15	18836.5
2021-03-19	32.79	38.35	96.79	21.96	18854
2021-03-22	32.95	38.73	95.61	21.8	18815.1
2021-03-23	32.95	38.66	92.45	20.94	18669.8
2021-03-24	33.48	38.08	91.57	20.86	18628.3
2021-03-25	33.46	38.34	92	20.89	18651.1
2021-03-26	33.67	38.49	91.87	20.75	18752.6
2021-03-29	33.87	39.55	91.25	20.53	18719.2
2021-03-30	33.44	39.43	94.23	20.03	18705.6
2021-03-31	33.1	39.01	93.11	20.59	18700.7

Figure 1: Daily stock prices from 03/11/2021 to 03/10/2026

Date	ENB	ATD	MG	CCO	TSX	R_ENB	R_ATD	R_MG	R_CCO	R_TSX
2021-03-11	32.91	39.27	96.54	20.47	18844.6					
2021-03-12	32.85	39.83	96.76	21.25	18851.3	-0.00182	0.01426	0.00228	0.0381	0.00036
2021-03-15	32.99	39.76	97.41	22.92	18954.8	0.00426	-0.00176	0.00672	0.07859	0.00549
2021-03-16	32.79	40.03	95.12	22.97	18874	-0.00606	0.00679	-0.02351	0.00218	-0.00426
2021-03-17	32.87	39.71	97.67	23.61	18983.1	0.00244	-0.00799	0.02681	0.02786	0.00578
2021-03-18	32.43	38.89	95.94	22.15	18836.5	-0.01339	-0.02065	-0.01771	-0.06184	-0.00772
2021-03-19	32.79	38.35	96.79	21.96	18854	0.0111	-0.01389	0.00886	-0.00858	0.00093
2021-03-22	32.95	38.73	95.61	21.8	18815.1	0.00488	0.00991	-0.01219	-0.00729	-0.00206

Figure 2: Daily returns for ENB, ATD, MG, CCO, and TSX

where $i \in \{ENB, ATD, MG, CCO, TSX\}$. The computed daily returns (in percentage) are shown in Figure 2.

To get the annual expected return as an input of CAPM, I multiply the daily return by approximately 252 trading days:

$$R_{annual} = R_i \times 252 \tag{2}$$

The calculated annual returns are shown in Figure 4.

1c. Betas

Beta β of stock i is computed by:

$$\beta_i = \frac{\text{Cov}(R_i, R_m)}{\text{Var}(R_m)} \tag{3}$$

In Excel, the covariance between a stock and the market index was calculated by the following built-in functions:

- Covariance: =COVARIANCE.S(stock returns,index returns).
- Annualized Standard deviation: =STDEV.S(returns)*SQRT(252)

The calculated betas are shown in Figure 4.

1d. Volatility-return graph

The graph is shown as in Figure 3.

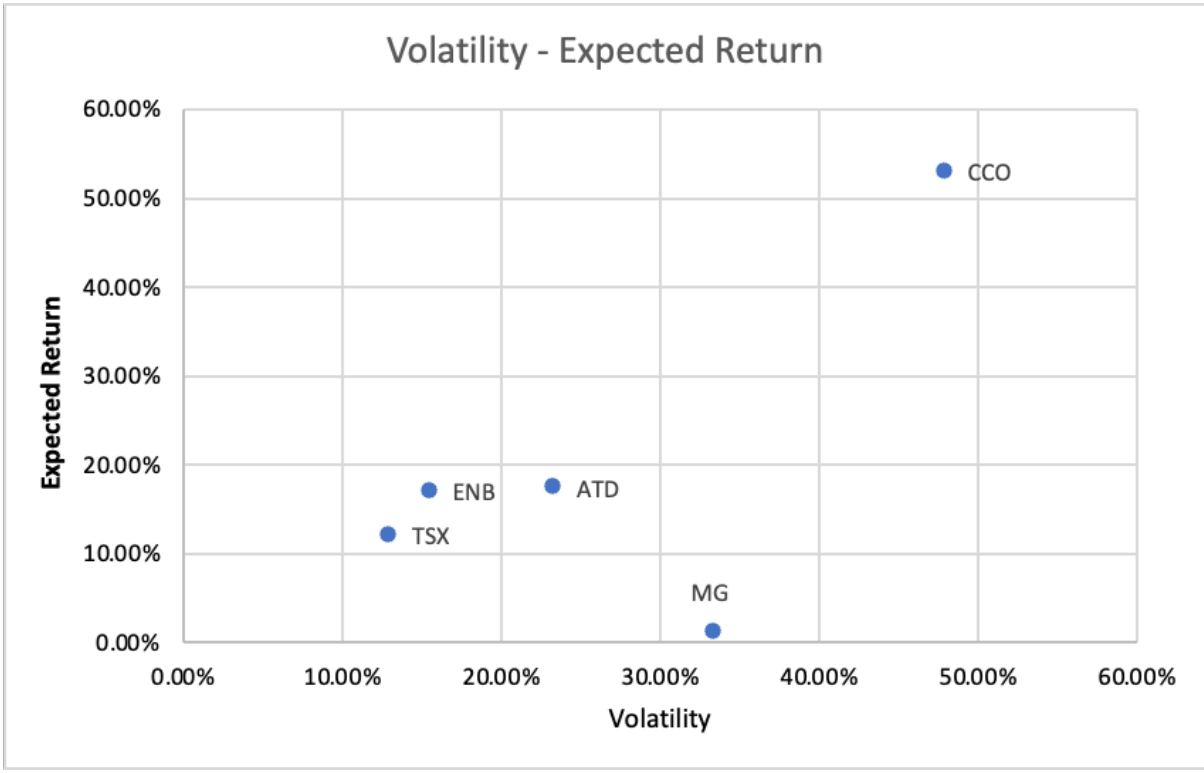

Figure 3: Volatility-Expected return graph

	ENB	ATD	MG	COO	TSX	
Covariance	0.0042%	0.0047%	0.0089%	0.0118%	0.0066%	
Realized (Annual) Expected return	17.21%	17.57%	1.39%	53.09%	12.27%	
Volatility	15.43%	23.24%	33.32%	47.84%	12.91%	
Beta	0.63	0.71	1.35	1.79	1	
Market risk premium						9.24%
CAPM Cost of Capital	8.89%	9.63%	15.46%	19.57%		
risk free rate						3.03%

Figure 4: Betas and Cost of Capital

1e. CAPM Cost of capital

I used the CAPM equation from textbook (Eq. 12.1, p.446, [3]). The computed cost of equity capital for each stock (security) is

$$E[R_i] = r_f + \beta_i \big(E[R_{\text{Mkt}}] - r_f\big) \tag{4}$$

- Risk-free rate $r_f = 3.05\%$ from Canada 5-year bond yield
- Expected return from market portfolio is 12.27%.
- Market risk premium is $E[R_{\text{Mkt}}] - r_f = 12.27\% - 3.05\% = 9.22\%$

Because r_f is annual bond yield, $E[R_{Mkt}](12.27\%)$ was computed as daily average $\times$ 252, and β is a unitless coefficient, the expected return of security i, $E[R_i]$, is expressed in APR without extra conversion. The calculated costs of capital are shown in Figure 4.

1f. Intuition of CAPM, DDM and DCFM comparison

CAPM's intuition

The CAPM states that an investor is compensated for bearing systematic risk which cannot be diversified away. So instead of investing on Government bond with lower return, investors are willing to take on additional risk and expect to earn an additional reward from their stocks. And since in a well-diversified portfolio, firm-specific risk cancels out, therefore only the market risk remains, which is exactly what CAPM depends on to compute the predicted return of securities.

The higher the β is, the higher required returns are. Under CAPM, the calculations have proved that: CCO (β = 1.79) has the highest cost of capital (19.57%), followed by MG (β = 1.35) with 15.46%, ATD (β = 0.71) with 9.63%, and ENB (β = 0.63) with the lowest return of 8.89%.

A notable observation is MG's realized return is 1.39% which is faw below the CAPM predicted return. It seems that MG's realized return underperfomed its required return over last 5 years, but it does not necessarily continues so in the future, as returns can change with market conditions and investor expectations.

DDM and DCFM comparison

- DDM estimates the cost of equity based on the present value of future dividends, which is capture by today stock priced. This is useful for firms with stable dividend payment, but sensitive to growth assumption and firms with unpredicted dividend growth.
- DCFM estimates the cost of equity based on the present value of future cash flows, which is the total cash the firm generate regardless of whether it pays dividends. The pros is the model values entire firm, both equity and debt, but it relies on many assumptions about the firm's cash flows (revenue, costs, etc.).
- CAPM estimates the cost of equity purely based on historical market data, thus does not relay on the firm's dividends or cash flows. However, CAPM models makes assumptions on all investors hold efficient portfolios, have same expectations, and can borrow or lend at the risk-free rate.

Overall, each models has its own trade-offs with their assumptions.

5 Question 2

2a. Correlation and Covariance matrices

In Excel, I use the following buil-in functions to obtain the correlation and covariance matrices between all 4 stocks:

- Covariance: =COVARIANCE.S($returns_i, returns_j$)
- Correlation: = CORREL($returns_i, returns_j$)

Both matrices are shown in Figure 5.

2b. Portfolio 1

My portfolio uses equal weights of 25% for ENB, ATD, CCO and MG.

Denote each stock by index $i \in \{1, 2, 3, 4\}$ where 1=ENB, 2=ATD, 3=CCO, and 4=MG.

Portfolio's expected return

The expected return of a portfolio is the weighted average of the expected returns of its stocks:

$$E[R_p] = \sum_i w_i \, E[R_i] \qquad \text{(Eq. 11.3, [3])} \tag{5}$$

- $E[R_p]$ = daily expected return of the portfolio
- $E[R_i]$ = expected daily return of stock i, estimated using historical daily return data

I multiply the daily expected return is then annualized by multiplying by 252 trading days to obtain the annual expected return (in percentage terms):

$$E[R_p]_{\text{annual}} = E[R_p]_{\text{daily}} \times 252 \tag{6}$$

Portfolio's volatility

To compute the portfolio's volatility, both the individual stock variances and the covariances between each pair of stocks are computed by the formula:

$$\begin{aligned}\text{Var}(R_p) = {} & w_1^2\text{Var}(R_1) + w_2^2\text{Var}(R_2) + w_3^2\text{Var}(R_3) + w_4^2\text{Var}(R_4) \\ & + 2w_1w_2\text{Cov}(R_1, R_2) + 2w_1w_3\text{Cov}(R_1, R_3) + 2w_1w_4\text{Cov}(R_1, R_4) \\ & + 2w_2w_3\text{Cov}(R_2, R_3) + 2w_2w_4\text{Cov}(R_2, R_4) \\ & + 2w_3w_4\text{Cov}(R_3, R_4)\end{aligned} \qquad \text{(Eq. 11.8, [3])} \tag{7}$$

$$\text{SD}(R_p)_{\text{daily}} = \sqrt{\text{Var}(R_p)_{\text{daily}}} \tag{8}$$

I scale the daily standard deviation by the square root of 252 to get the annual volatility:

$$\text{SD}(R_p)_{\text{annual}} = \text{SD}(R_p)_{\text{daily}} \times \sqrt{252} \tag{9}$$

In Excel, I used built-in functions to compute those above portfolio metrics, taking advantage of covariance matric, array of stock weights, and computed data from question 1:

- Portfolio return: =MMULT(returns vector,weights vector) * 252
- Portfolio volatility: =POWER(MMULT(MMULT(TRANSPOSE(I7:I10),C7:F10),I7:I10),0.5) * POWER(252, 0.5)

All calculated results are shown in Fugure 5

2e. Diversification benefits and drawbacks from hsitorical data

The diversification benefit results in the lower portfolio's volatility (20%) compared to the weighted average of individual stock volatilities. This happens because the portfolio has low pairwise correlations among 4 stocks (ranging from 0.15 to 0.21), so their returns do not move closely together. The combination of stocks helps to diversify the idiosyncratic risk, leaving the portfolio exposed to only systematic risk.

	Canada 5 year bond (11 March, 2026)	3.03%			
		Returns ENB	Returns ATD	Returns MG	Returns CCO
	Expected daily returns	0.068%	0.070%	0.006%	0.211%
a)	Covariance	Returns ENB	Returns ATD	Returns MG	Returns CCO
	Returns ENB	0.00009452	0.00003144	0.00004448	0.00005976
	Returns ATD	0.00003144	0.00021440	0.00006029	0.00006757
	Returns MG	0.00004448	0.00006029	0.00044065	0.00012191
	Return CCO	0.00005976	0.00006757	0.00012191	0.00090823
	Correlation	Returns ENB	Returns ATD	Returns MG	Returns CCO
	Returns ENB	1	0.220881702	0.217937292	0.203968548
	Returns ATD	0.220881702	1	0.196144547	0.153119708
	Returns MG	0.217937292	0.196144547	1	
	Returns CCO	0.203968548	0.153119708	0.192698963	1
b)	Portfolio daily expected returns	0.09%			
	Portfolio daily StDev	1.23%			
	Portfolio annual expected returns	**22%**			
	Portfolio annual StDev	**20%**			
c)	Portfolio beta	1.120990204			

Name	Weights
ENB	25%
ATD	25%
MG	25%
CCO	25%
Sum	**100%**

Figure 5: Portfolio 1

However, we are computing risks and returns from historical data, which has limitations. Firm strategies, industry conditions, and macro shocks can all change the relationships among stocks, i.e. the current pattern in their movements may not repeat in the future. Therefore we should combine the computed return and risk under CAPM model with more market/firm metrics to have a more accurate prediction.

2f. Frontier Efficient curve (Bonus)

To construct the efficient frontier, I used Excel Solver to find the portfolio allocation that minimizes variance for a given target return. I set the portfolio's expected return to a fixed level and gave constrains of sum of weights to be 1, and all weights are at least 0. Solver then adjusted the weights to minimize the portfolio volatility.

This process was repeated for a range of target returns. From computed expected return of each individual stocks from question 1, I got the highest return with CCO (53.09%) and the lowest return with MG (1.39%). Therefore, my target returns is the set $\{2\%, 6\%, 10\%, ..., 46\%, 52\%\}$. I then plotted the returns and volatilities of optimal portfolios obtained to get the Efficient curve:

By definition, the efficient frontier is the set of portfolios that offer the highest expected return for a given level of risk. A notable observation is ATD receives zero weight throughout the frontier as it is dominated by ENB, a stock with similar return but lower risk. Besides, MG also gets zero weights at high target return ranges, as it has very high risk (33.32%) compared to very low return it has (1.39%), so easily pushed out from the portfolio.

Another notable point is the minimum variance portfolio (MVP) consists entirely of ENB (100%). By Equation 7, with positive correlations, the more the stocks move together and the higher their covariance or correlation, the more variable the portfolio will be. In my portfolio, ENB already has the lowest volatility (15.43%), but the volatility spread is too large (from 8-32%) relatively to other stocks such that no correlations among stocks can improves the diversification benefit. Therefore, ENB dominate all and the MVP only holds ENB with volatility of 12.04% and return of 17.21%.

The corner portfolio is highly visible because of the abruption that ENB dominates over other stocks. And for both upper and lower segments from the efficient frontier, I only have a combination of 2 stocks in the portfolio, either ENB+MG or ENB+CCO. As a result, the curvature is unclear and the line segments look veyr linear. The curve is shown in Figure 6.

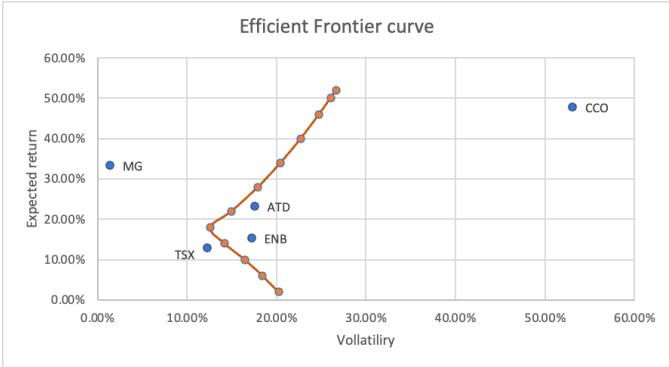

Figure 6: Efficient Frontier Curve

	Canada 5 year bond (11 March, 2026)	3.03%			
		Returns ENB	Returns ATD	Returns MG	Returns CCO
	Expected daily returns	0.068%	0.070%	0.006%	0.211%
		Returns ENB	Returns ATD	Returns MG	Return CCO
	Returns ENB	0.00009452	0.00003144	0.00004448	0.00005976
	Returns ATD	0.00003144	0.00021440	0.00006029	0.00006757
	Returns MG	0.00004448	0.00006029	0.00044065	0.00012191
	Return CCO	0.00005976	0.00006757	0.00012191	0.00090823
	Portfolio daily expected returns	0.10%			
	Portfolio daily StDev	1.09%			
	Portfolio annual expected returns	0.257321129			
	Portfolio annual StDev	0.172736205			
a)	**Tangent portfolio's Sharpe ratio**	**1.314264886**			
b)	**Tangent portfolio's Beta**	**0.922269899**			
	TSX's Sharpe ratio	0.72			
	TSX's annual returns	12.27%			
	TSX's volatility	12.91%			
c)	return differences	13.46%			
	volatility differences	4.36%			

Name	Weights
ENB	57%
ATD	20%
MG	0%
CCO	24%
Sum	100%

Figure 7: Efficient Frontier Curve

6 Question 3

3a. Sharpe ratio maximized by Solver

Since short-selling is not permitted, I use Solver in Excel to find the tangent portfolio by maximizing the Sharpe ratio with constraint of weight sum to be 1, and each weight cannot be negative.

The maximzed Sharpe ratio is 1.31 (by Solver). This is the highest return per unit of risk that can be obtained.

The tangent portfolio has ENB 57%, ATD 20%, CCO 24%, and MG 0%. MG receives zero weight because its realized return of 1.39% falls below the risk-free rate of 3.03%, generating a highly negative excess return. Therefore, MG will contributed negatively to the Sharpe ratio and is excluded from the optimal risky portfolio. Sharpe ratio is calculated by:

$$\text{Sharpe Ratio} = \frac{E[R_P] - r_f}{\text{SD}(R_P)} = \frac{E[R_P] - 3.03\%}{\text{SD}(R_P)} \qquad (Eq11.7, [3]) \tag{10}$$

The result is shown in Figure 7.

3b. Tangent portfolio's Beta

I used built-in Excel functions to compute the beta of the tangent portfolio using the optimal weights solved by Solver. The result is shown in Figure 7.

The tangent portfolio's beta of 0.922 which is below the market systematic risk ($\beta = 1$), meaning the portfolio is less volatile than the market portfolio (TSX index).

3c. Tangent Portfolio vs Index

The tangent portfolio has higher volatility than the TSX (by approximately 4.36 percentage points), but gets much higher return (by 13.46 percentage points). This suggest that the tangent portfolio offers a better risk-return trade-off. The tangent portfolio has Sharpe ratio of 1.31 compared to beta of TSX of 0.72, proving that it rewards higher return at any given level of risk. The following calculations are to compute the differences in returns and volatility between tangent portfolio and TSX as mentioned above:

1. Difference in (expected) returns:

$$\Delta E[R] = E[R_{\text{tangent}}] - E[R_{\text{TSX}}] = 25.73\% - 12.27\% = 13.46\% \tag{11}$$

2. Difference in volatility:

$$\Delta SD = SD_{\text{tangent}} - SD_{\text{TSX}} = 17.27\% - 12.91\% = 4.36\% \tag{12}$$

The graph is shown in Figure 8.

An investor can always combine the tangent portfolio with the risk-free asset along the Capital Market Line to achieve any desired volatility level. In fact, based on the CML formula, any risky portfolio combined with risk-free assets on the CML has the volatility as a fraction of the volatility of the tangent portfolio.

For example, if investors want the portfolio of combination of risky and risk-free assets to have the same volatility level as of TSX at 12.91%, we can compute the invested fraction x in the risky assets by using formula (Eq. 11.16, [3]):

$$\begin{aligned} SD(R_{xP}) &= \sqrt{(1-x)^2 \operatorname{Var}(r_f) + x^2 \operatorname{Var}(R_P) + 2(1-x)x \operatorname{Cov}(r_f, R_P)} \\ &= \sqrt{x^2 \operatorname{Var}(R_P)} \\ &= x\, SD(R_P) \end{aligned} \tag{13}$$

$$x = \frac{SD(R_P)}{SD(R_{xP})} = \frac{SD_{(}R_{\text{portfolio}})}{SD_{(}R_{\text{tangent portfolio}})} = \frac{12.91\%}{14.05\%} = 0.92 \tag{14}$$

3d. Tangent portfolio vs Portfolio 1 (equal weights)

The three portfolios can be summarized in the following table:

Portfolio 1 vs Tangent portfolio

My portfolio 1 has equal weight of 25% for each stock and achieves a return of 22% with a volatility of 20% and a Sharpe ratio of 0.949. Clearly that the equal weighting has dragged

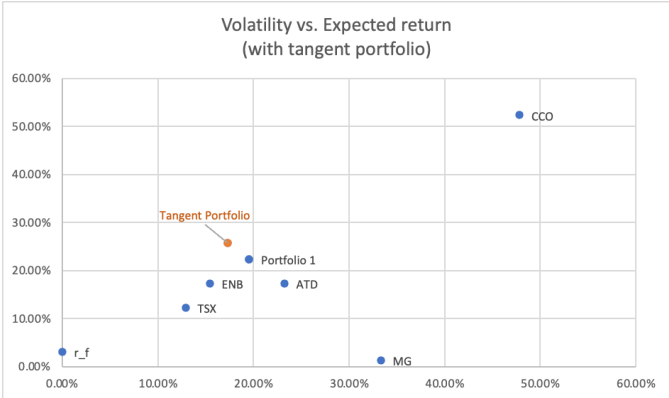

Figure 8: Volatility vs. Expected return (with tangent portfolio)

	Portfolio 1 (Q2)	Tangent Portfolio (Q3)	TSX Index (Q1)
Weights	25% each	57/20/0/24%	Market-cap
Annual Return	22%	25.73%	12.27%
Annual StDev	20%	17.27%	12.91%
Beta	1.121	0.922	1.00
Sharpe Ratio	0.949	1.314	0.716

Table 1: Portfolio 1, TSX, Tangent portfolio

down the portfolio's Sharpe ratio to due to MG's high volatility and negative excess return.

Portfolio 1 vs Index

Portfolio 1 also has better return compared to TSX, but at significantly higher volatility (20% vs 12.91%) and higher beta (1.121 vs 1.00). It implies that my Portfolio 1 takes on the highest systematic risk because of its highest volatility, but its return is still lower than the tangent portfolio, so diversification can still be improved by reweighting the contained stocks.

7 Question 4

4a. CAPM line plot

I plotted the expected returns and volatility of all the securities in the original portfolio, tangent portfolio, portfolio 1, TSX index, and risk free rate. The CAPM line is the straight orange line that connects the risk free rate with the tangent portfolio, as shown in Figure 9.

4b. SML plot

The Beta-Return graph is shown below with the SML in red line, as shown in Figure **??**.

4c. Deviations from SML

No. All the dots are not on the red line. The formula for the deviation from SML (measured by alpha) is:

$$E[R_i] = r_f + \beta_i \left(E[R_{\text{Mkt}}] - r_f\right) \qquad (Eq.12.6, [3]) \tag{15}$$

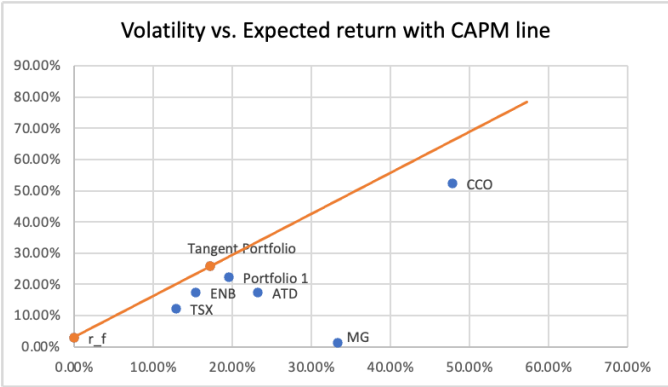

Figure 9: CAPM line

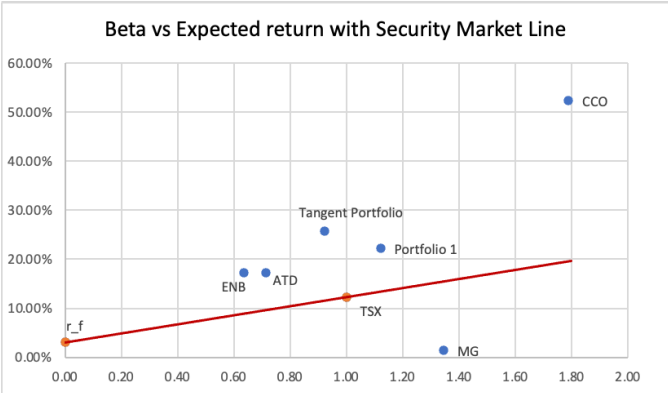

Figure 10: Security Market Line

The intuition is that CAPM implies the only risk that gets rewarded is systematic risk, which is measured by β, because idiosyncratic (firm-specific) risk can always be diversified away by adding more and more securities to the portfolio.

Therefore, SML has positive, upward sloping, so the higher β is (i.e. the higher risk sensitivity to the market), the higher expected return is. And if securities lie on SML, it means that CAPM predicts $\alpha = 0$ for them.

Non-zero α occurs because SML is constructed using expected returns (forward-looking), while we are plotting the realized returns from historical stock price data (in past 5 years). As a result, as realized returns have already reflected firm's idiosyncratic shocks, they cause all the stocks to deviate from CAPM predictions, either above or below the SML.

From the plot, clearly MG has the most deviation from SML. Its strongly negative alpha ($\alpha = 15.46\% - 1.39\% = -14.07\%$) implies that investors are bearing systematic risk but not being compensated for. Its β = 1.35 says it should earn 15.46%, but it only earned 1.39%. Either MG had a lot of bad idiosyncratic shocks during the sample period (2021 - 2026), or the market isn't efficient due to rational behaviours (discussed more in 4d).

4d. Assumptions and empirical justification of the SML

I will discuss about 3 key assumptions of CAPM (as mentioned in [3]).

1. Perfect Capital Markets: from plot of Beta versus Expected return, all securities lie above or below SML, indicating that alpha is non-zero. This happens because the real markets are not frictionless, and transaction costs, taxed, or short-selling constraints are present.

2. Investors choose Efficient Portfolios: Any investor, even uninformed ones, can hold the

TSX, which is efficient, to get zero alpha, thus no rational investor should accept $\alpha < 0$. However, from the plot, some investors are holding MG despite its negative alpha instead of holding TSX. This only happens when some of them:

- **Overconfidence bias**: In my portfolio, despite MG has $\alpha \approx -14\%$, some investors have been holding it. They must either misinterpret the information and believe MG will bounce back, or they care about something beyond return (eg. they might have **disposition effect** if they work at MG, and refuse to sell a loser).
- **Care about things beyond return and volatility**: some just want to buy stocks from firms they are familiar with (**familiarity bias**), or care about how well the portfolio performs compared to their peers.

3. Homogenous expectations: CAPM assumes all investors agree on expected returns, volatilities, and correlations for all securities. In practice, this is clearly too strong as investors differ in information received, investing skills, and financial resources. However, it can be relaxed by only assume that investors hold rational expectations (p. 487, [3]). My results show that ENB ($\alpha \approx 8.4\%$), ATD ($\alpha \approx 7.9\%$), CCO ($\alpha \approx 34\%$), and MG ($\alpha \approx -14\%$), all deviating from the SML.

   As discussed in Section 2e, my data sample only has 5 years of daily returns. The average realized return is only an estimate of the expected return and could be very noisy over a short time period, leading to estimator error. Therefore, the realized returns likely reflect idiosyncratic shocks during the sample period, not what investors expected when they chose to buy the stock. The SML, however, reflects expected returns implied by CAPM.

   Last but not least, market might be inefficient temporarily, because competition will drive alpha toward zero over time. If ENB and CCO's positive alphas were persistent, rational investors would buy them, pushing the pices up and alpha is pushed down to 0. In other words, the non-zero alphas might likely to reflect a temporary deviation, or due to estimation error in my daily returns data in last 5 years.

8 Financial Advice

Conclusion from Analysis

First, I concluded that stock selection fundamentally is crucial for maximizing returns while trying to get a good diversification. In my portfolio, the weakest stock is Magna International (MG), despite a very big automotive supplier, only having a realized return of 1.39% over the sample period, falling below the risk-free rate of 3.03% and generating a negative alpha. Under CAPM, investors are holding more risks but less returns for compensation, so MG is not a good addition for a well-diversified portfolio.

Secondly, weighting chosen stocks is very important to get an optimal portfolio. It is not always good to randomize weight selection, or give highest weight to the stock with highest return. Risk-return tradeoff analysis as well as other metrics such as Sharpe ratio need to be considered for portfolio optimization.

Finally, the computed tangent portfolio (Sharpe = 1.314) is actually better than TSX index, implying that TSX index is not the efficient portfolio predicted by CAPM theory. CAPM is an useful model to approximate the reality, but it should not be the only tool used to assess a portfolio.

Caveats and Further Considerations

The data shows that stocks deviates from the SML: MG with negative alpha, while ENB and ATD have positive alpha. The positive alpha trading strategies does not automatically imply that investors can earn free profits without risk. Instead, we should look at (1) whether the market portfolio is truly efficient and (2) how risk is measured.

A weak point about measurement problem is the **proxy error**, where TSX is just a proxy for Canada market, but the true market includes bonds, real estate, human capital, etc. So beta might be estimated from an inefficient benchmark, not real mispricings, causing stocks to have non-alphas even if the true market portfolio is efficient.

Investors can have **behavioural biases** which they might favor familiar or big, popular stocks, or show disposition effect. These biases distort the demand, but because total demand must equal total supply, sophisticated investors are forced to take more small, value, momentum stocks. Therefore, alpha opportunities appear because the market portfolio is distorted by biased investors, not because there is free profits.

There is also another way to think about risk besides measuring how portfolio moves with the market by beta. They are **non-tradable wealth** that investors face risk outside the market, and the major one is due to human capital (job income). For example, a banker who is tied to finance sector, already exposed to finance risk, will avoid finance stocks and want to buy other sectors to offset their job risk. Therefore many investors deviate from market portfolio. This distorts demand and can push stock away from SML.

Based on the above reasons, these deviations from SML do not necessary imply a non-valid CAPM, but can be expected in practice.

Recommendation

Based on the analysis, I recommend to consider the tangent portfolio (ENB 57%, ATD 20%, CCO 24%, and MG 0%) which offers the highest Sharpe ratio and it should be combined with the risk-free asset (as the tangent one has a fized risk level and different investors have different tolerance) to control how much risk to be taken.

MG should be excluded from the portfolio, as its return falls below the risk-free rate, and its highly negative alpha adds volatility that are not compensated. However, as mentioned from the analysis, all computed measures is based on data from past 5 years. If an investor has a reason to believe MG's future growth that differs from its past performance and firm's characteristics that are not captured by CAPM, they can consider a reasonable amount of MG stock.

Finnaly, any portfolio should be reviewed regularly. Positive alphas or any arbitrage opportunities are easily to taken advantaged quickly by other investors, and the portfolio today might be no longer optimal in a short future, because news, expectations, and global events are constantly changing.

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